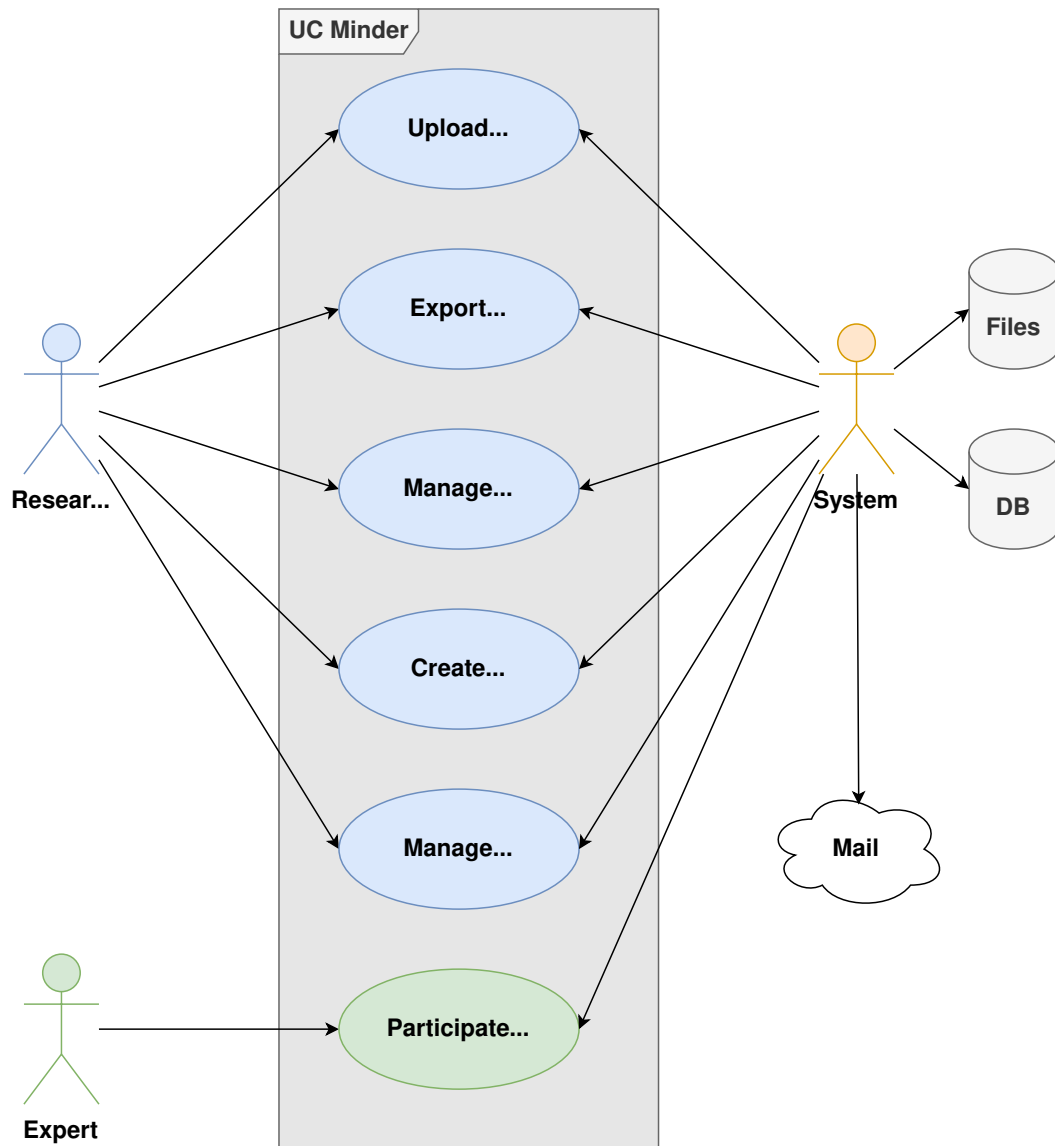


Minder - Tinder Style Data Labeling

Requirements Analysis

We analyzed the special use cases and assume a standard user management system, providing functions like password resets.

Use Case Diagram



Use Cases

1	Manage Users
Description	The researcher creates, manages and deletes users
Actors	Researcher
Preconditions	The researcher is authenticated. Emails can be sent
Result	Researchers or experts can be managed
Trigger	e.g. Data has to be labeled
Process	1. The researcher navigates to “Users” 2. The researcher creates a user with an email 3. The researcher assigns the user a role 4. The researcher can delete the user
Requirements	Emails have to be delivered within a minute
2	Upload Data
Description	The researcher uploads a dataset without labels
Actors	Researcher
Preconditions	The researcher is authenticated
Result	Named datasets can be used for studies
Trigger	Data without labels
Process	1. The researcher navigates to “Data” 2. The researcher creates a named dataset 3. The researcher uploads single files or archives into this dataset
Requirements	A storage for a large amount of files
3	Create study
Description	The researcher creates a study
Actors	Researcher
Preconditions	The researcher is authenticated and a dataset exists
Result	Experts are notified about a new study
Trigger	A new question arises
Process	1. The researcher navigates to “Studies” 2. The researcher creates a study from a dataset, labels are configured 3. The researcher publishes the study for all experts(?)
Requirements	-

4	Manage study
Description	The researcher changes or ends a study
Actors	Researcher
Preconditions	The researcher is authenticated and a study exists
Result	Anonymous access and corrected mistakes
Trigger	Sharing the study or fixing a mistake
Process	1. The researcher navigates to “Studies” 2. The researcher selects a study 3. The study can be shared with a QR code 4. All configured options can be changed 5. The study can be ended 6. The study can be deleted
Requirements	-

5	Participate in Study
Description	Experts label data
Actors	Experts
Preconditions	The expert is authenticated or using a QR code and part of a study
Result	Data is labeled
Trigger	A new study was published
Process	1. The expert selects a study from his home page 2. The expert labels a singular data point 3. The expert can keep on playing the same study 4. The expert can close the study or navigate back
Requirements	A random data point is selected from all non labeled data for the expert

6	Export data
Description	The researcher exports data
Actors	Researcher
Preconditions	The researcher is authenticated and a study exists
Result	Exported data
Trigger	Data has been collected for a study
Process	1. The researcher navigates to “Studies” 2. The researcher selects a study with data 3. The researcher exports the data and downloads it
Requirements	-

Questions

- How are study participants selected? Are all registered users allowed to participate?
- What is the data import and export format? Should there be a batch upload?
- How does the QR code map to an anonymous user? Is the anonymous user shared? Should each QR code visit get a unique id?
- What exactly is meant by “Different amount of cells per experiment”?

Ideas

- Different UI hints for labels, colors instead of text
- Help menu during a study / experiment
- Redoing a study / known data for error correction / What about bias

Gamification / Motivation

- Percentage of completion
- Statistics and user profile / community biases and opinions
- Achievements (studies participated in, time spent, veteran, ...) and medals for continued usage
- Counter for labeled data points / global High-Score
- News section to follow up on studies and how the data was useful